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Research on Movement Intentions of Human's Left and Right Legs Based on Electro-Encephalogram Signals

Active rehabilitation can use electro-encephalogram (EEG) signals to identify the patient's left and right leg movement intentions for rehabilitation training, which helps stroke patients recover better and faster. However, the lower limb rehabilitation robot based on EEG has low recognition accuracy so far. A classification method based on EEG signals of motor imagery is proposed to enable patients to accurately control their left and right legs. Firstly, aiming at the unstable characteristics of EEG signals, an experimental protocol of motor imagery was constructed based on multijoint trajectory planning motion of left and right legs. The signals with time-frequency analysis and event-related desynchrony/synchronization (ERD/S) analysis have proved the reliability and validity of the collected EEG signals. Then, the EEG signals generated by the protocol were preprocessed and common space pattern (CSP) was used to extract their features. Support vector machine (SVM) and linear discriminant analysis (LDA) are adapted and their accuracy of classification results are compared. Finally, on the basis of the proposed classifier with excellent performance, the classifier is used in the active control strategy of the lower limb rehabilitation robot, and the average accuracy of the left leg and right leg controlled by two healthy volunteers was 95.7%, 97.3%, 94.9%, and 94.6%, respectively, by using the ten-fold cross test. This research provides a good theoretical basis for the realization and application of brain-computer interfaces in rehabilitation training. [DOI: 10.1115/1.4055435]

Keywords: EEG, movement intentions, active training control, rehabilitation training

1 Introduction

Stroke disease has a high incidence in the elderly population [1]. About three-quarters of stroke patients will be left with sequelae, such as upper and lower limb mobility, language communication difficulties, defecation difficulties, and more than two-thirds of stroke patients live with hemiplegia [2–3]. The lower limb rehabilitation training method for these patients can be divided into passive training and active training [4]. The passive training method doesn't involve the active consciousness of patients [5,6]. Conversely, active training method is that the patient's active motion intention is collected to accurately control the auxiliary rehabilitation training equipment. It can not only make the affected limb get effective gait stimulation training, but also reshape the neural circuits of the patient's motor brain area. Meanwhile, the active training method can speed up the nervous system's control of the muscles [7]. At present, patients' active intention is mainly recognized based on physiological signals of the human body, including electromyograpghy (EMG) signal, biomechanical signal and electro-encephalogram (EEG) signal, and so on [8–11]. However, for early stroke patients with severe limb motor dysfunction, the application of EMG signal and biomechanical signal is limited to a certain extent. However, bypassing the peripheral nerve and musculoskeletal system, using EEG signals to directly extract neural activity has gradually become a research hotspot.

The brain-computer interface technology serves as a bridge between the human brain and control devices and the collected EEG signals are classified and translated into instructions that the computer can execute [12]. These instructions can realize direct interaction and control between the brain of patients with limb motor dysfunction and external devices [13–16]. EEG can be divided into evoked EEG and spontaneous EEG. Evoked EEG signals mainly contain P300 (A type of event related potential (ERP)) and steady-state visual evoked potential (SSVEP). It is an involuntary passive stimulus that allows users to selectively focus their attention on visual stimuli. The spontaneous EEG signals mainly refer to MI (motor imagination). Motor imagery generates different potentials by imagining the movements of the body to output control instructions without external stimulation [17,18].

At present, the recognition of evoked and spontaneous EEG signals has been studied at home and abroad. Studies based on evoked EEG, such as the highly popular P300 spellers based on visual or auditory paradigms [19,20], realized six kinds of brain signals' controlling of lower limb exoskeleton by SSVEP [21]. Research on EEG recognition based on motor imagery has been carried out, such as wheelchair control [22], robot arm control [23], mouse movement control [24], and so on. Due to the short distance between left and right leg motion-related brain regions of the head, it is more difficult to identify the left and right legs compared with the human's upper limbs. Gao et al. [25] proposed a leg prosthesis control method based on motor imagination with EEG signals, which triggered motor imagination through video cues, and used C3, C4, and Cz channels to collect signals. Then, support vector machine (SVM) classifier and directed acyclic graph (DAG) structure were combined to classify three kinds of virtual tasks (up, down, and walk), and the classification accuracy is 81.11%. Gordleevaal et al. [26] proposed a method to control the exoskeleton of lower limbs based on motion imagination and trigger motion imagination through computer prompts. The 8-lead EEG cap (FCz, C5, C3, C1, Cz, and C26) was used for signal acquisition. The method used spatial filtering to identify features and used linear discriminant classifier to classify the modes of controlling lower limbs with three commands (left hand, right hand, and rest). The average accuracy of the method was about

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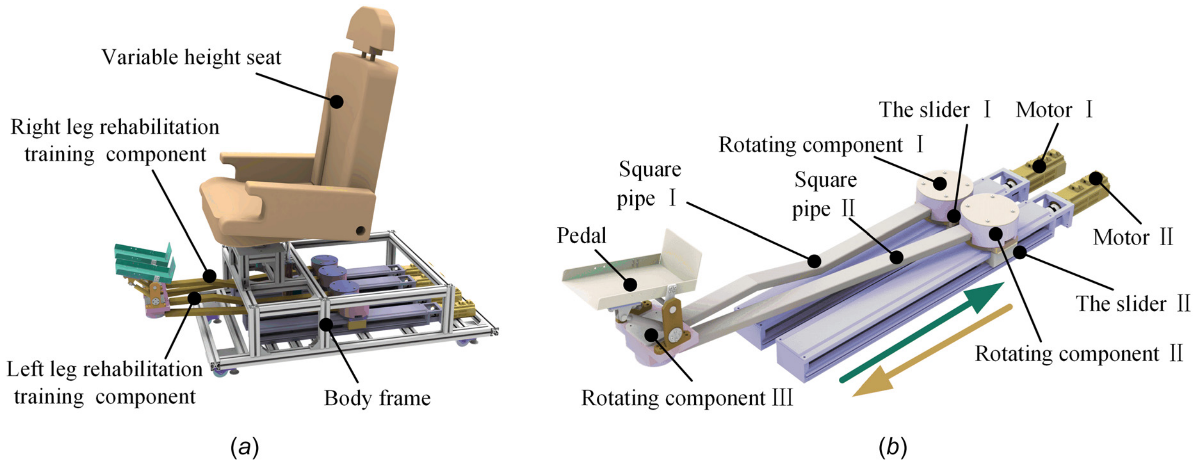

Fig. 1 Mechanical design of lower limb rehabilitation robot: (*a*) overall structure of lower limb rehabilitation robot and (*b*) mechanical structure of right lower limb rehabilitation training component

70%. Kwak et al. [27] developed an asynchronous brain–machine interface-based lower limb exoskeleton control system based on SSVEPs. SSVEPs consist of five light-emitting diodes fixed to the exoskeleton. Using eight electrodes (PO3, PO, PO4, PO7, O1, OZ, O2, PO8) EEG cap sampling signal. KNN algorithm was used to recognize five movements of lower limbs (walk forward, turn right, turn left, sit, and stand), and the final recognition accuracy was about 91.3%.

Due to the low signal-to-noise ratio of EEG signal, poor signal generalization, and differences in attention and physiology information between subjects, the accuracy of lower limb recognition based on EEG signal is still not very high. Aiming at making the coordinated rehabilitation training movements of humans' left and right legs, how to generate EEG signals, and how to extract and classify the effective features of the collected EEG signals have become important research content to accurately identify the intention of stroke lower limb movement to control external rehabilitation robots.

2 Method

2.1 Module Construction of Lower Limb Rehabilitation Robot. The overall mechanical device of the lower limb rehabilitation robot is shown in Fig. 1(*a*), which is mainly composed of variable height seat, body frame, left leg rehabilitation training component, and right leg rehabilitation training component. The left leg rehabilitation training component and the right leg rehabilitation training component are symmetrical. The mechanical structure of the right leg rehabilitation training component is shown in Fig. 1(*b*). The rotating component I and rotating component II have installed on slider I and slider II, respectively. The axis of rotating component I and rotating component II are perpendicular to the moving plane of the right leg rehabilitation training component. Meanwhile, rotating component I and rotating component II are fixed with one end of square pipe I and square pipe II, respectively. The other ends of the square pipe I and square pipe II are connected through a rotating component III. A pedal is fixed on the rotating component III. Under the drive of motor I and motor II, slider I and slider II can realize forward and backward movement. Then, based on the difference between the two motors' velocities, the two degree-of-freedom plane motion of the pedal can be realized. The patient's foot will be fastened on the pedal to achieve planar movement, so the hip and knee joints of the patient's left and right legs can realize the couple motion in the human sagittal plane and coronal plane.

2.2 Construction of Electro-Encephalogram Signal Collected System Based on Motor Imagery. As shown in Fig. 2, the control system is mainly composed of EEG cap, EEG amplifier, synchronization marker, PC I and PC II. PC I and PC II are used for data collection, experimental protocol playing, data transmission, and communication. Their work is divided into offline training and online training (the solid line represents the offline training process, and the dashed line represents the online training process). Offline training process includes: Firstly, the designed experimental canonical form is designed and used as visual stimulus, and the subjects are asked to watch the experimental protocol videos on PC I at rest for motor imagery. A 64-channel EEG cap is adopted for recognition and induction of EEG signals. Then the collected signals' type is marked by synchronous marker. Finally, the EEG amplifier processes the collected signals and stores them on PC II. After the offline experiment, all the stored data are processed and analyzed, and a classifier model is trained. Online training process includes: Firstly, the subject used an off-line training paradigm to perform motor imagery during the left leg and the right leg kick separately. This will generate new EEG data. Then the new EEG data are fed into the offline trained classifier model in real-time for classification and recognition. Finally, the result of recognition is corresponding to the coded action to drive the lower limb rehabilitation robot.

2.3 Design of Experimental Canonical Form. It is necessary to design the experimental protocol reasonably. Visual stimulus is used to help humans generate the corresponding motor imagery action. Figure 3 is the design of experimental protocol, in which subjects are required to imagine the movement of the left leg or the right leg. Each subject is required to complete two groups of experiments (one group of right leg experiments and one group of left leg experiments). Each group of experiments had three rounds in total, and each round had 20 motion imagination experiments. In other words, each group of subjects was required to complete 60 motion imagination experiments based on random hints. The time sequence diagram of single motion imagination is shown in Fig. 3. Each experiment lasts for 10 s. "+" appears in the center of the screen from 0 s to 3 s, prompting the subjects to pay attention; then the sentence "Observation" appears on the screen from 3 s to 5 s, prompting the subjects to imagine; Later, the left/right leg movement video randomly appears on the screen from 5 s to 10 s. The subjects carried out corresponding motion imagination according to the content of the video (as shown in Fig. 3, the first group was the movement of the right leg and the second group was the movement of the left leg). There will be a 5 min rest in the middle of each round to prevent fatigue of the subjects from affecting the quality of EEG signals. The total length of offline training is about 20 min.

2.4 Electro-Encephalogram Signal Acquisition and Preprocess Method. Saga64/32 is used in the experiment which is manufactured by Hernan TMSI. As a noninvasive EEG device, it

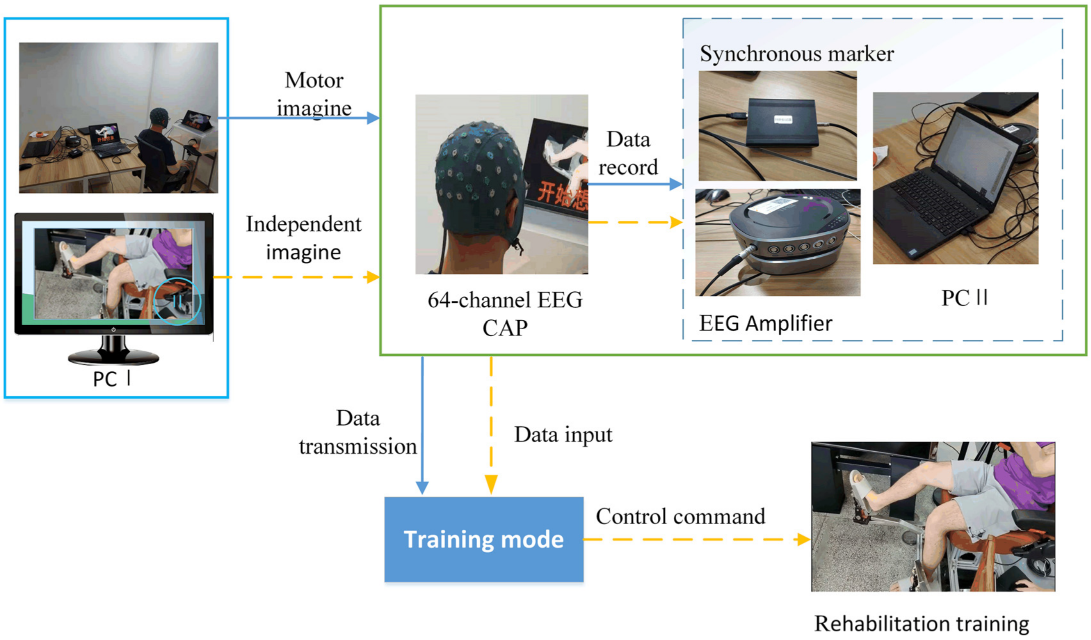

Fig. 2 EEG signal collected system based on motor imagery

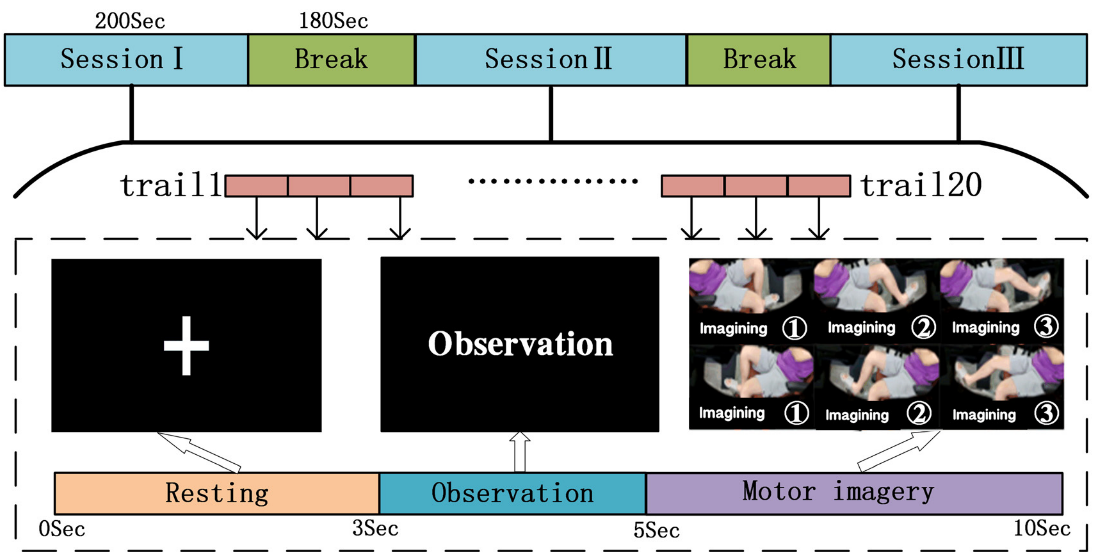

Fig. 3 Experimental protocol design

can provide high-quality data and an EEG testing system for a wide range of application scenarios. The EEG cap has 32 channels of active electrodes. Electrode distribution position according to 10–20 international system positioning [28] (C1, C2, C3, C4, CP1, CP2, CP5, CP6, CPz, Cz, F3, F4, F7, F8, FC1, FC2, FC5, FC6, FCz, FP1, FP2, Fz, O1, O2, P3, P4, P7, P8, Pz, T7, and T8). The reference electrode and the ground electrode are placed on the left and right earlobes, respectively. The process of EEG data acquisition and preprocessing is shown in Fig. 4. Firstly, the sampling frequency is set at 500 Hz during the experiment, and the impedance is reduced to less than 30 K Ohms by injecting conductive paste into the EEG cap before EEG signal is collected. The working principle is to measure the potential difference between the active electrode on the EEG cap and the reference electrode, and then carry out analog-to-digital conversion and filtering. Finally, the weak signal is processed through the amplifier and the amplitude of the signal is convenient for data analysis.

The methods of preprocessing the collected signals will be introduced below. Due to a long experiment time, the conductive performance of the paste injected into the EEG cap will fail, leading to some abnormal or lost data. The first step of preprocessing is to check the data by setting the maximum and minimum thresholds (250 Hz, 0 Hz) [29]. When the data exceeds the threshold, it is regarded as abnormal data. It can be corrected by using interpolation method, which uses the average of the data of the three channels around the abnormal data channel to correct the data. The second part of preprocessing is denoising the data. Because the target signal is mainly low-frequency and the power line interference generally brings 50–60 Hz noise. So a certain degree of denoising can be achieved by using a low pass filter (the cutoff frequency is 30 Hz). In addition, EEG signal is a kind of nonrhythmic signal, and it is mixed with other signals (such as EMG signal, electro-oculography signal, etc.) so that these signals and EEG signals have the overlap of space–time cross. In EEGLAB toolkit, the independent principal component analysis (ICA), which is a method of looking for potential factors or components in multidimensional statistics, was used to identify EEG, electro-oculography, and EMG artifacts. The components separated by

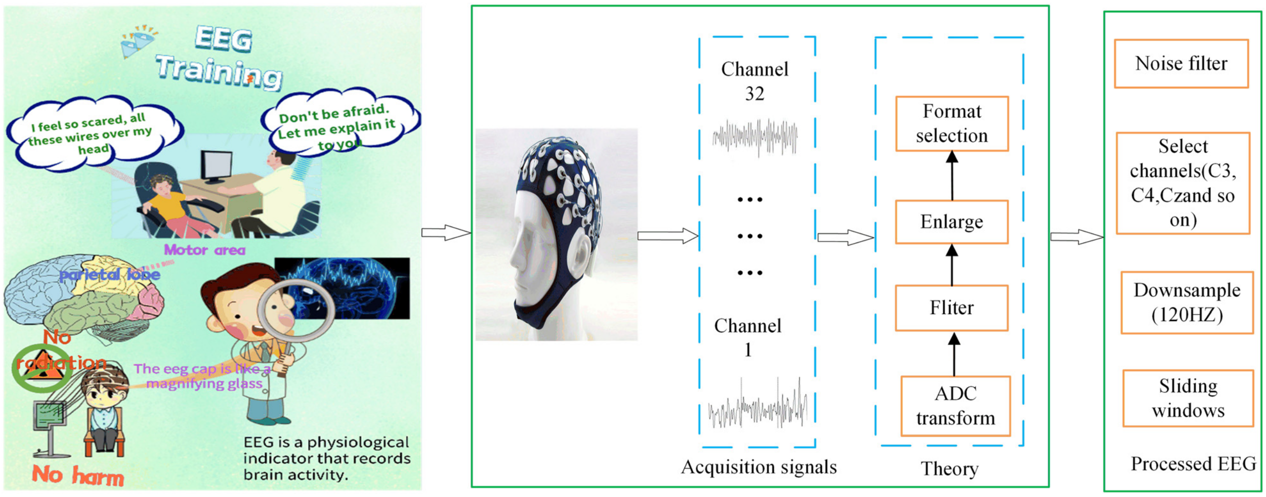

Fig. 4 Flow chart of EEG signal acquisition and preprocess

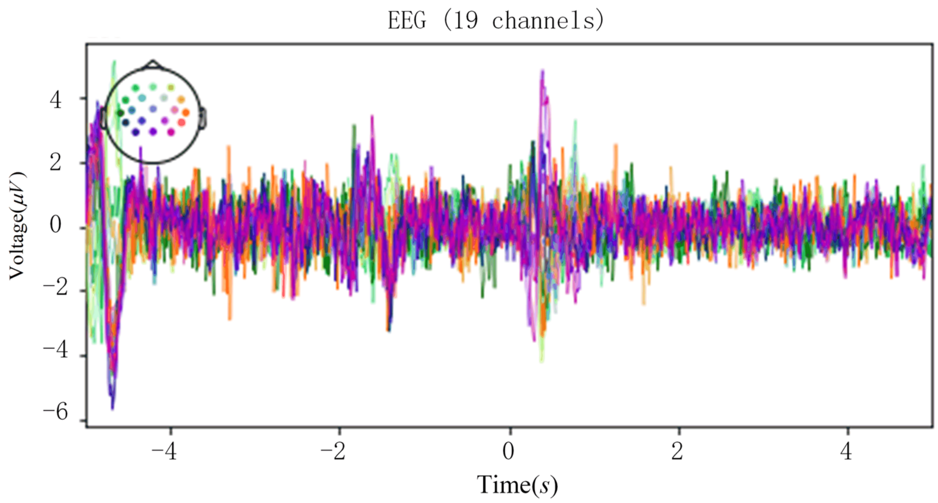

Fig. 5 ERP waveform during the experiment

ICA are manually removed to complete the denoising process. In the last step of preprocessing, the frequency can be reduced to 120 Hz by using down-sampling method to improve the data processing speed. Finally, the sliding window technology (The sliding window time length is set to 150 ms, and the overlap rate is set to 50%) is adopted to improve the overall data sample size.

Data were collected and preprocessed according to the experimental protocol, data acquisition principle, and preprocessing process. At present, the main reason is that the EEG experiment is not popular, and the subjects and patients do not understand the EEG experiment. So they develop feelings of fear. The volunteers in this experiment were two healthy students with no history of neurological diseases, lower limb pathology or gait abnormalities. Before the experiment, we will give reasonable and scientific explanations to the subjects through interesting and understandable posters, so that the subjects can correctly understand the standardization and safety of EEG experiment. This can minimize the patients' worries and fears about the experiment, so as to ensure the effectiveness of the experimental data source. Their ages are 22 and 24, respectively. This experiment has been approved by the Ethics Committee of School of Mechanical Engineering, Ningbo University (Approval number: BE2051), and healthy volunteers have been informed and signed the informed consent for the experiment. Before the experiment, the experiment content and the behavior corresponding to the screen display instructions were explained to the subjects in detail, and a 2-minute pre-experiment was conducted.

2.5 Reliability Analysis of Electro-Encephalogram Signals. The validation of EEG signals includes the validation of experimental canonical form, the evaluation of subjects' motor imagination ability, and the reliability analysis of the experiment. ERP is an event-related potential, which can reflect the changes of neuro electrophysiology in the brain during the cognitive process, and can present the time-domain waveform under a specific frequency band. Because the EEG signals are susceptible to noise, they are measured differently each time. Firstly, the collected data were preprocessed. The 19 channels over the motor functional areas (C1, C2, C3, C4, CP1, CP2, CP5, CP6, CPz, Cz, FC1, FC2, FC5, FC6, FCz, O1, O2, P3, P4, P7, P8, Pz, T7, and T8) were selected as the data of the ERP analysis. The data of each channel needs to be normalized uniformly. The left and right legs of both volunteers were tested according to the experimental canonical form for 10 s. Figure 5 shows the EEG of one volunteer during an experiment which is after all the data are preprocessed and then superimposed on average. The ERP waveform was obvious when the motion imagination began at 0 s. In this experiment, the generation of ERP was stimulated by different visual tasks, so different distributions of ERP waves appeared at −5 s, −2 s, and 0 s, which was consistent with the visual stimulus designed in the experimental canonical form. Finally, it is proved that the design of experimental paradigm is reasonable.

The research is a rehabilitation training for the motor imagination ability of the subjects, which needs to verify the motor ability of the subjects to ensure the effectiveness of the signal. Time-frequency analysis was conducted on the data collected by the above method. Figure 6 shows the time-frequency analysis of the left and right legs of subject 1. The brain activity caused by the motion imagination of the left and right legs changed significantly at electrode positions C3, C4, and CZ. Therefore, packages of MNE can be used (mne.time_frequency.tfr_multitaper) to conduct time-frequency analysis on data of C3, C4, and CZ channels, the principle of which was to calculate time-frequency representation by using DPSS cone. Important parameters in the code were set as freqs = np.logspace (*np.log10([2,30]), num = 20). Figure 6(*a*) shows the time-frequency analysis of the left leg. The energy at CZ at 10 Hz and 20 Hz is about 0Au and −5Au, respectively, one second before the imagination (−1 s in the figure). Start by imagining that in the second after (1 s in the figure), the energy at CZ at 10 Hz and 20 Hz is about −8Au and −12Au. As shown in Fig. 6(*b*), one second before the experimental paradigm 5 s (4 s in the figure), the energy at CZ at 10 Hz and 20 Hz is about 0 Au and −3Au. One second after 5 s of the experimental paradigm (6 s in the figure), the energy at CZ at 10 Hz and 20 Hz is about −8Au and −9Au. Therefore, it can be concluded that the energy of the motor area of the brain decreased significantly before and after 0 s, and the subjects would have an obvious ERD phenomenon when imagining the movement of the left and right legs. It is shown that the subjects have certain imagination ability of left and right leg movement, which meets the basic conditions of rehabilitation training and can ensure the validity of EEG data.

On the other hand, the actual time for rehabilitation training of left and right legs is 5 s in the process of EEG signal acquisition. Due to the visual delay of motor imagery, only 3 s before motor imagery is extracted and analyzed in data processing and analysis. Power spectral density (PSD) is commonly used to analyze the changes in the power characteristics of the signal frequency domain in different states. It can be used to observe the event-related desynchronization and synchronization phenomenon (ERD/S) of the EEG signal in the process of motor imagination. In the ideal motion imagination experiment, the ERD phenomenon shows the power reduction in the specific frequency band (8–13 Hz mu wave) under the state of motion imagination in PSD analysis [30]. Figure 7 shows the PSD of two volunteers' imagining left/right leg movement. The blue line, red line and green line, respectively, represent the mean value of PSD of all channels in the three stages of rest, preparation, and beginning of imagination. The shaded part represents the mean value plus or minus the standard deviation of PSD. The dispersion degree of PSD value in each stage can be analyzed through the shaded part.

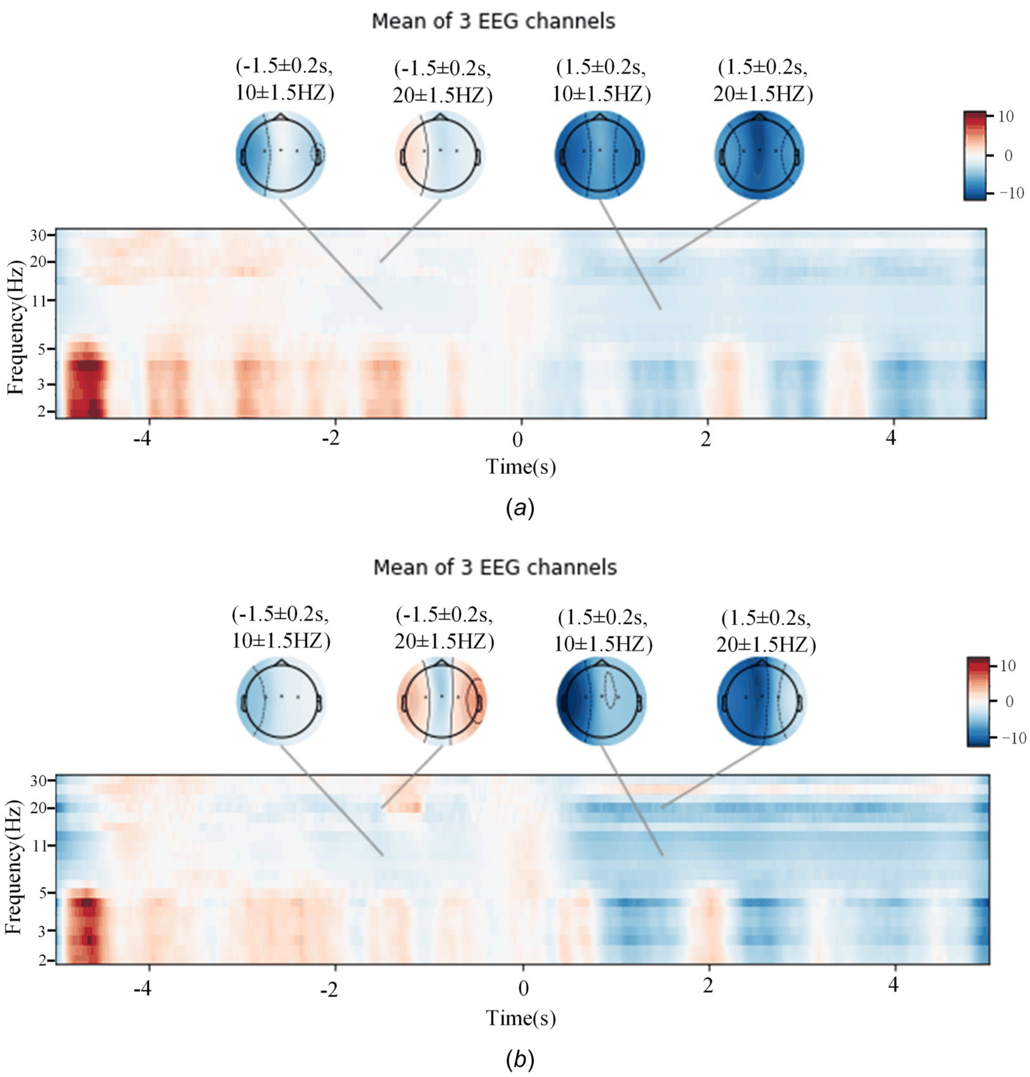

Fig. 6 Time-frequency analysis of subjects: (*a*) time-frequency analysis of the left leg and (*b*) time-frequency analysis of the right leg

It can be seen that the PSD of the green line (5 s–8 s motor imagery phase) is lower than that of the blue line (rest) and red line (ready imagery) in the band range of around 10 Hz. Therefore, the design of the experimental paradigm has obvious ERD phenomenon in the volunteers' leg motion imagination stage, which shows the reliability of the proposed experiment.

2.6 Feature Selection of Electro-Encephalogram Signals. Feature extraction of EEG data is required, which can enhance the variance between EEG classes, improve the recognition effect of EEG signals and improve the accuracy of classifier. EEG signals can be extracted by different methods and different classifiers according to different tasks. The commonly used feature extraction methods in literature include PSD, CSP, wavelet transform (DWT), and so on. EEG signals are spontaneously generated through motor imagery, which can cause an increase or decrease in neural activity, called event-related synchronization or desynchronization (ERS/D). In order to distinguish ERS/D, CSP has been widely used, because of its ability to maximize variance differences between two different categories. Therefore, CSP may be the best method for feature extraction [31]. The feature extraction method of CSP is to find a set of optimal projection Spatial filters by using matrix diagonalization. It can maximize the variance of one type of signal and minimize the variance of another type of signal. The principle is to obtain the feature vector with high identification and extract the spatial distribution characteristics of each signal. Therefore, CSP method is used to extract features from 32-channel EEG data. The specific methods of CSP feature extraction are as follows.

Let $\overline{X_l}$ and $\overline{X_r}$ be the EEG data matrices of the left and right legs' movement based on motion imagination, respectively, whose dimensions is $a*b$. Where a represents the total number of channels of EEG data and b represents the sample number of each channel (sampling frequency * motor imagery time). The normalized spatial covariance of left and right legs' EEG signals can be calculated as follows:

$$\begin{cases} C_1 = \dfrac{\overline{X_l}\,\overline{X_l}^T}{\operatorname{trace}\left(\overline{X_l}\,\overline{X_l}^T\right)} \\[2ex] C_r = \dfrac{\overline{X_r}\,\overline{X_r}^T}{\operatorname{trace}\left(\overline{X_r}\,\overline{X_r}^T\right)} \end{cases} \tag{1}$$

The mean normalized covariance matrices C_l and C_r are obtained by averaging all experiments in each group. C_l is the expectation of the spatial covariance matrix of the first type of sample data (left leg's movement based on motion imagination), and C_r is the expectation of the spatial covariance matrix of the second type of sample data (right leg's movement based on motion imagination). $\overline{X_l}\,\overline{X_l}^T$ represents the trace. The sum of spatial covariance matrix of imaginary data of left and right leg movement can be expressed as

$$C = C_1 + C_r = V_0 \lambda V_0^T \tag{2}$$

where V_0 is the eigenvector matrix of the matrix λ, λ is the diagonal matrix of the corresponding eigenvalues. The whitening

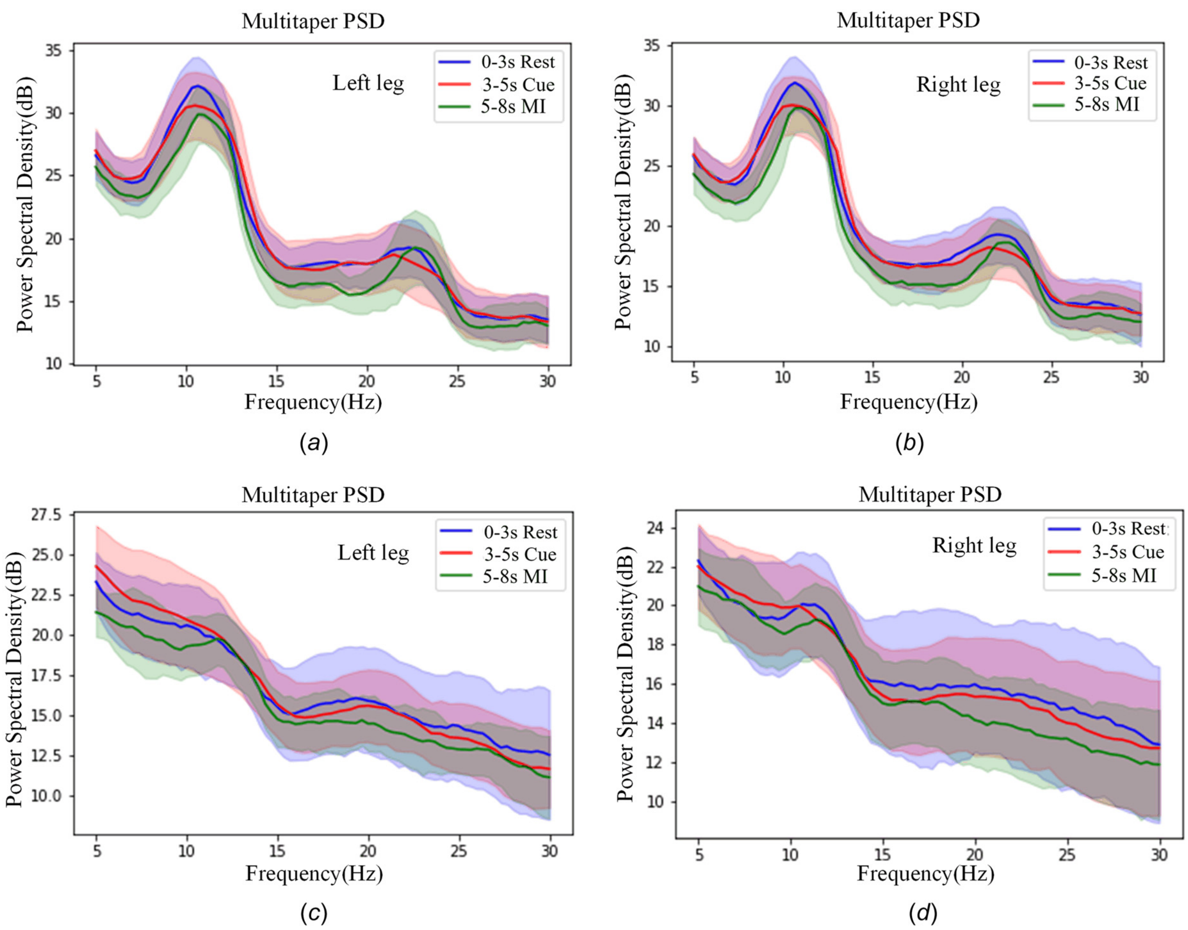

Fig. 7 Variation diagram of multitaper PSD value of subject 1 and subject 2 in three different stages of motor imagination: (a) multitaper PSD of subject 1's left leg movement, (b) Multitaper PSD of subject 1's right leg movement, (c) Multitaper PSD of subject 2's left leg movement, and (d) Multitaper PSD of subject 2's right leg movement

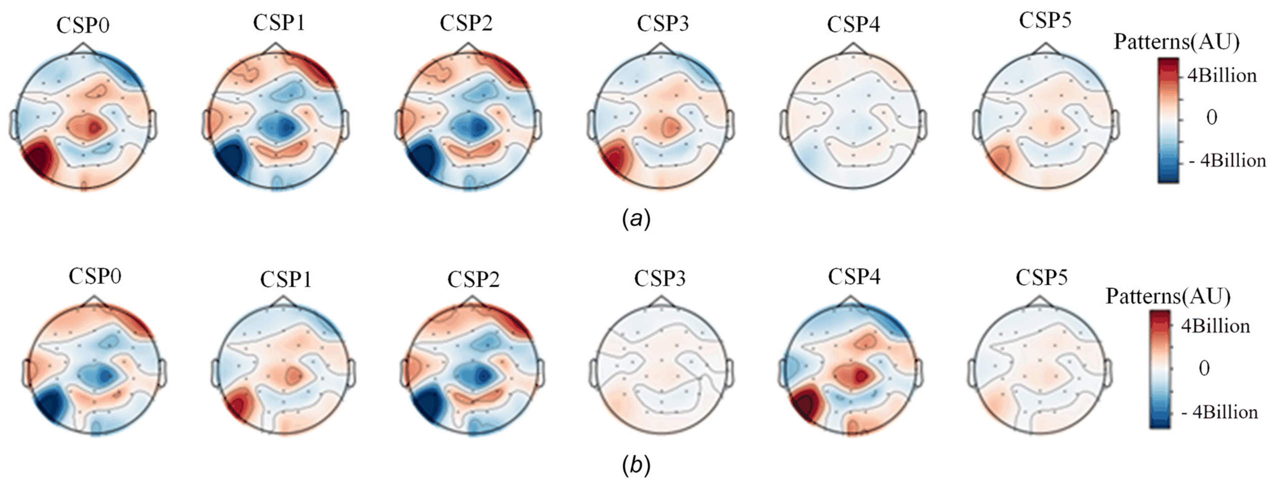

Fig. 8 Left and right leg motion imagination: (*a*) right leg motion imagination and (*b*) left leg motion imagination

Table 1 The coding labels of the left and right leg training mode

Training mode	MI Type	Encoding
Right leg training mode	Right leg	1
	Rest	0
Left leg training mode	Left leg	−1
	Rest	0

transformation matrix P is transformed to the average normalized covariance as follows:

$$\begin{cases} p = \sqrt{\lambda^{-1}} V_0{}^T \\ S_l = PC_l P^T \\ S_r = PC_r P^T \end{cases} \tag{3}$$

For S_r and S_l, the eigenvector matrices S_r and S_l of matrices V_1 and V_2 are equal, and the sum of the diagonal matrices λ_1 and λ_2 of the two eigenvalues is the identity matrix

$$\begin{cases} S_l = V_1 \lambda_1 V_1{}^T \\ S_r = V_2 \lambda_2 V_2{}^T \\ V_1 = V_2 \\ \lambda_1 + \lambda_2 = I \end{cases} \tag{4}$$

The eigenvectors with the largest eigenvalues correspond to one category, and the eigenvectors with the smallest eigenvalues correspond to the other category. For separating the variance in the two signal matrices, the whitening transformation of the eigenvector from the original EEG data $\boldsymbol{W}$ to the maximum eigenvalue is optimal. The projection matrix can be calculated as $W = V^T P$, where $\boldsymbol{W}$ is the projection matrix. The raw EEG data is multiplied by the projection matrix $\boldsymbol{W}$, which converts the raw EEG into unrelated components. The original EEG can be reconstructed by multiplying the reciprocal of $\boldsymbol{W}$ by $\boldsymbol{Z}$

$$\begin{cases} Z = WX \\ Z^* = W^{-1} Z \end{cases} \tag{5}$$

where Z is the source component of the EEG, including the common and specific components of different tasks; The columns of W^{-1} are called spatial modes and are the distribution vectors of brain power sources.

The first column of W^{-1} shows the maximum difference for one task and the last column shows the minimum difference for the other task. Because visual stimuli in the experimental paradigm design activated 3 s and all samples in the CSP activation time window were used. Therefore, CSP chose the 3 s time window and used all the sample data. As shown in Fig. 8, the CSP method was used to extract features from the data of the left and right imaginary legs of volunteer 1, and then MNE was used to visualize the extracted features. When CSP is used for feature extraction visualization, the first four feature phenomena are more obvious in the right leg motion imagination, and the first five feature phenomena are more obvious in the left leg motion imagination. Therefore, the minimum parameter of CSP feature extraction is set as 4. It is good to see a significant change of energy in the brain during motor imagery. The optimal parameter selection will be verified in the following section.

2.7 Selection of Classification Algorithm and Determination of Characteristic Parameters. Electro-Encephalogram signals need to be classified after signal acquisition, preprocessing, and feature extraction. In order to identify the motion intention of humans, the choice of classifier has a great influence on the recognition accuracy. The common methods of EEG recognition with higher accuracy experiment and then the classifier with higher accuracy is selected by comparison. SVM is a special traditional machine learning algorithm [32], which seeks to minimize the structural risk to improve the generalization ability of learning machine, and to minimize the experience risk and confidence range. The algorithm can obtain good statistical rules under the condition of the minimum statistical sample size. Linear discriminant analysis (LDA) is often used to reduce the dimension of binary classification and multiclassification [33]. For the two categories of data, it achieves feature extraction (dimension reduction) by projecting the high-dimensional sample data into the optimal discriminant vector space. After projection, the sample data can be guaranteed to have the maximum class distance and the minimum class distance in the new subspace, which means that it has the best separability in the subspace.

Under the design of SAGA device in experimental protocol, the dimension of collected signal is $60 \times 32 \times 750$, and the data dimension after feature extraction is $(60+60) \times 19 \times 250$. The coding labels of the right leg training mode and the coding labels of the left leg training mode are shown in Table 1. Then, the amount of data is expanded through the operation of a sliding window. After processing, the amount of data is changed to $1200 \times 19 \times 125$, which is the data of the input classifier.

The data were collected by two volunteers in offline training mode, and the dataset was divided into training set and test set,

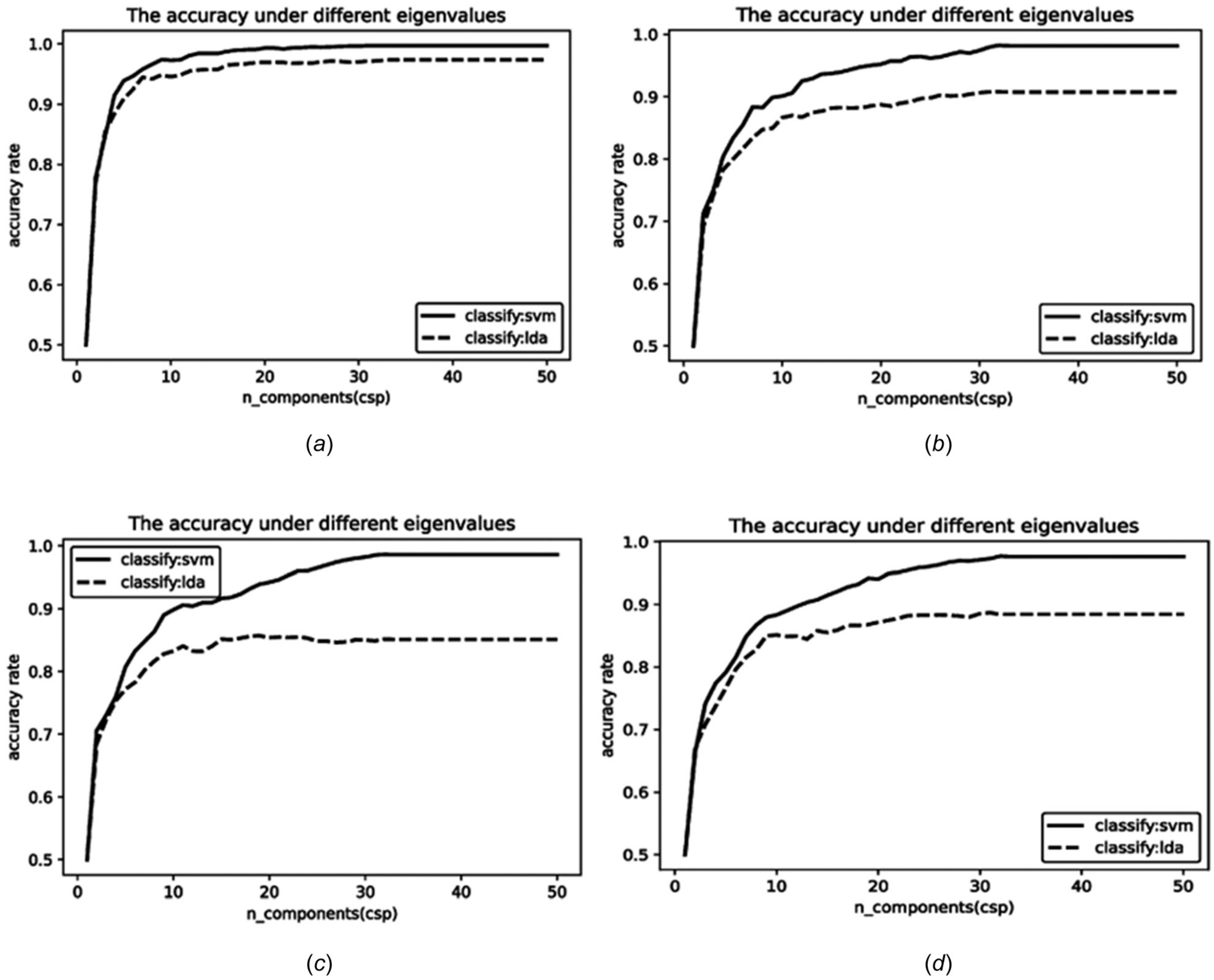

Fig. 9 The accuracy of two subjects' motion imagination under different algorithms: (*a*) subject 1 in right leg training mode, (*b*) subject 1 in left leg training mode, (*c*) subject 2 in right leg training mode, and (*d*) subject 2 in left leg training mode

accounting for 80% and 20%, respectively. Then, SVM and LDA classifiers were used to identify the motion intention of the left and right legs. The classification accuracy of different classifiers for different motion imaginations of the two volunteers was shown in Fig. 9. As shown in Fig. 9(*a*), n_components is a parameter of the CSP algorithm. When volunteer 1 was in the right leg training mode and the number of n_compinents was 7, the classification effect of SVM was better than that of LDA classifier. When n_components is 32, the accuracy of SVM is up to 99.70%. In the left leg training mode, as shown in Fig. 9(*b*), the classification effect of SVM is also better than that of LDA. When the parameter n_components is 33, the classification accuracy of SVM reaches the maximum 98.12%.

As shown in Fig. 9(*c*) when volunteer 2 was training his right leg, the classification accuracy of SVM was better than that of LDA, and when parameter n_components was 33, the accuracy could reach the highest level of 98.58%. In the left leg training mode, as shown in Fig. 9(*d*), the classification accuracy of SVM is better than that of LDA, and when parameter n_components is 33, the accuracy can reach the highest level of 97.62%. Therefore, SVM is chosen as the classifier for the system recognition.

In brief, during the right leg training mode, the system can identify the humans' motor imagination intention with 99.14% average accuracy. During the left leg training mode, the average accuracy of the system was 97.87%. In the feature extraction process with CSP method, the parameter n_components is set 33, and the SVM classification algorithm has the highest accuracy in identifying the imaginary intention of left and right leg movement.

3 Electro-Encephalogram Trigger Scheme Design and Experimental Verification

3.1 Electro-Encephalogram Trigger Scheme Design. During the online training, the left and right leg multijoint coupling system based on EEG signal was adopted. In the training mode, when the subject imagines the movement of the left or right legs,

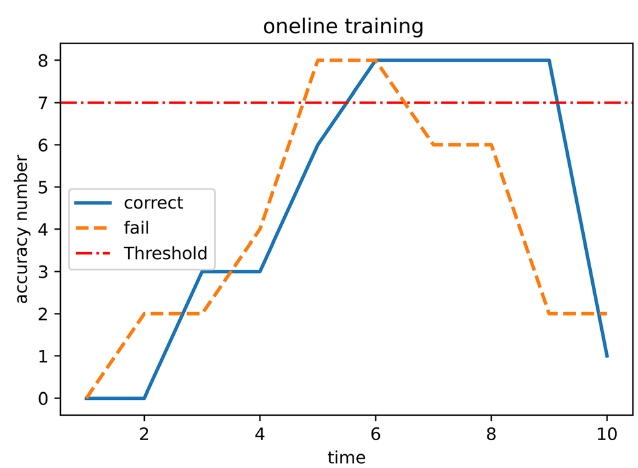

Fig. 10 Successful triggering strategies for motor imagery

Table 2 10 fold cross-validation of subject 1

	1	2	3	4	5	6	7	8	9	10	Average
Right_leg_model	0.966	0.979	0.987	0.975	0.987	0.995	0.954	0.95	0.966	0.970	0.973
Left_leg_model	0.970	0.929	0.970	0.95	0.979	0.958	0.929	0.95	0.966	0.966	0.957

Table 3 10 fold cross-validation of subject 2

	1	2	3	4	5	6	7	8	9	10	Average
Right_leg_model	0.966	0.966	0.954	0.937	0.941	0.908	0.95	0.937	0.954	0.975	0.946
Left_leg_model	0.941	0.954	0.962	0.979	0.975	0.945	0.891	0.95	0.962	0.929	0.949

the lower limb rehabilitation robot will be triggered to conduct movement training for the corresponding left or right legs. The triggering scheme was that 10 epochs were taken as the input data of the model during 3 s of motion imagination, and the model judged 10 epochs in 3S. The initial threshold is set as 7. When 7 or more are correctly identified in 3 s, the output is 1 or −1; otherwise, it is 0. In the online real-time identification of the left and right leg training mode, if the output of the control algorithm is 1 or −1, the lower limb movements of the corresponding training mode in the online training can be started according to the corresponding hard-coded instructions. As shown in Fig. 10, if the motion imagination of 5 s–8 s is successfully identified more than 7 times within 3 s, it is regarded as successful imagination. Solid line corresponds to imagination success, dashed line corresponds to imagination failure.

3.2 Electro-Encephalogram Control Experiment. After the establishment of the lower limb rehabilitation robot system based on BCI control, experimental verification of its control system is needed to ensure that humans' active consciousness can be accurately reflected and fed back during rehabilitation training. Therefore, experimental verification of the BCI control system is carried out. The k-fold cross-validation method can better ensure the model in the system without over-fitting phenomenon and make the generalization of the model get certain improvement and guarantee. According to the size of the total amount of data, 10-fold cross validation is selected. The data were divided into 10 groups, and 10% of the data were extracted from each group for identification and judgment with the trained model. The average value of the 10 groups of judgment test results was used as the final verification result. Tables 2 and 3 show the verification test results of the two subjects. Subject1 will have an average validation accuracy of 97.3% for right-leg training and 95.70% for left-leg training. Subject2 will have an average validation accuracy of 94.6% for right-leg training and 94.9% for left-leg training. On the whole, the average recognition accuracy of motor imagination intention was 95.95% in the right leg training mode, and 95.3% in the left leg. The high accuracy of the identification of left and right legs in online training indicates that the system has reliability in active rehabilitation training.

4 Discussion

In summary, a lower limb rehabilitation robot based on human left and right leg motion intention recognition is constructed which can accurately identify the intention of left and right lower limb movement. The validity of the experimental protocol is verified to ensure the reliability of the collected data. Two training modes of left and right legs were used to identify patients' motor imagery with high accuracy (left leg movement and rest, right leg movement and rest). At present, Gao et al. [25], Gordleeva et al. [26], and Kwak et al. [27]. have made advances in the field of lower-limb intention recognition. But there are some problems, such as low classification accuracy and influence on EEG signal caused by focusing attention on video. This study has made improvements to the above problems. EEG signals were generated by motor imagery. Parameters in CSP method were adjusted through experiments. SVM was used as classifier. The average accuracy of left and right leg training was 95.95%. After the experiment, the subjects filled in the experimental feedback questionnaire. The results of the questionnaire showed that the active rehabilitation training was safe and meaningful. At the same time, the experimental feedback content also verified the feasibility for patients.

There are still some limitations in this study, which need to be improved in the follow-up research. Firstly, aiming at the problem that EEG signals are vulnerable to the influence of environment and surrounding electromagnetic and the weak generalization ability of CSP in the experiment. Multiple sets of data can be collected to analyze the problem that the weak generalization ability of EEG signal feature extraction can be affected, which reduces the training time of patient model. Meanwhile, more experimental volunteers will be recruited to compare the recovery effect before and after rehabilitation training, so as to formulate rehabilitation training plans and time more scientifically. In addition, after extracting features using CSP method, it is difficult to classify the feature images directly by visualizing the CSP images imagined by the subjects in their left and right leg movements. In the following study, the application of deep learning image classification method will be studied to improve it and carry out multiclassification. It is also difficult to classify the feature images directly by visual CSP images after feature extraction using CSP method. In the following experiments, we will also study the application of deep learning image classification methods to improve it and carry out multiclassification. The high accuracy of recognition improves the initiative of patients' lower limb rehabilitation training and promotes the recovery of lower limb motor ability to a certain extent. In order to quantify the rehabilitation effect, patients will be followed for a long time in the future, and the biological indicators of lower limbs at the same stage will be quantified as part of the later work. In the future, this application scenario is not limited to rehabilitation medical robots. At the same time, it has a certain application prospect in the fields of home human-computer interaction games, human-computer cooperation in specific scenes, and unmanned driving.

5 Conclusion

A set of multijoint motion systems based on EEG control of motor imagery was built based on the above research and exploration. Aiming at the movement intention of the left and right legs, the multijoint rehabilitation movement module of the left and right legs was constructed, and the recognition method based on EEG signals of motor imagination was proposed to control and drive it. The validity and reliability of the design experimental paradigm were verified by time-frequency analysis and ERD/S analysis. For EEG signal extraction features, CSP method was used to maximize the variance difference between two different categories. At the same time, two commonly used classification algorithms (LDA and SVM) were used for experimental verification, and the SVM classifier was used and the n_components parameter in CSP extraction method was set to 32, so that the effect of recognition classification model was optimized. Finally, the control system based on EEG signals of motion imagination is applied to the multijoint motion coupling module of left and right legs. Experimental results show that the system has high recognition accuracy, and the feasibility of the proposed method is shown. The results show that the proposed method can control the lower limbs to a certain extent, which provides a theoretical basis for promoting the development of motor imagination in the field of rehabilitation.

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